

# Sounak Mondal

 [sounakcs.github.io](https://sounakcs.github.io) |  [somondal@cs.stonybrook.edu](mailto:somondal@cs.stonybrook.edu) |  [LinkedIn](#) |  [Google Scholar](#)

## RESEARCH INTERESTS

I am interested in research on **vision-language modeling**. My PhD thesis focuses on using **vision-language representation learning** and **multimodal foundation models (e.g., Multimodal LLM)** for modeling human visual attention (eye gaze).

## EDUCATION

### **Stony Brook University**, Stony Brook, NY

Doctor of Philosophy, Computer Science (**GPA: 4.0/4.0**)

Aug 2020 – June 2026

Advisors: Minh Hoai Nguyen, Dimitris Samaras, Gregory Zelinsky

### **Jadavpur University**, Kolkata, India

Bachelor of Engineering, Computer Science & Engineering (**GPA: 8.65/10, Rank: 3/54**)

Aug 2013 – May 2017

## EXPERIENCE

### **Meta**

**Research Scientist Intern**, Reality Labs Research, Burlingame, CA

Jun 2025 – Dec 2025

- Modeling Vision and Sensor Data via Multimodal LLMs.

**Research Scientist Intern**, Reality Labs Research, Redmond, WA

Jun 2024 – Dec 2024

- Vision-Language Modeling of Eye Gaze Behavior (published at ICCV 2025, patent filed)

### **CV Lab, Stony Brook University**

**Graduate Researcher**, Stony Brook, NY

Nov 2020 – Present

- Vision-Language Modeling for gaze prediction (CVPR'23, ECCV'24), and gaze decoding (ICCV'25 accepted + 1 under review)
- Gaze prediction modeling for search (ECCV'22, CVPR'24, CVPR'25); multimodal modeling for gaze estimation (ECCV'24)
- Modeling attentional states, e.g., mind-wandering vs. sustained attention, from eye gaze during online learning (ongoing)

### **Samsung**

**NLP Engineer**, Natural Language Processing Team, Bangalore, India

Jun 2017 – Aug 2020

- Enhancements of **Bixby** virtual assistant.
- Low resource intent classification via transfer learning
- Sequence labeling models for Named Entity Recognition and speech end-point detection
- Lightweight and fast text classification architecture (ICSC'20)

**Summer Intern**, SVoice Team, Bangalore, India

May 2016 – Jul 2016

- Context awareness in SVoice platform for Natural Language Processing

### **UII America**

**Research Intern**, Computer Vision, Cambridge, MA

May 2023 – Aug 2023

- Vision-Language Modeling: Scene Graph Generation from captions using Large Language Models (LLMs).

### **Indian Statistical Institute**

**Undergraduate Researcher**, Kolkata, India

Jul 2015 – Jun 2017

- Video Action Recognition/Detection (ICAPR'17, ICVGIP Workshop'16).

## SELECTED PUBLICATIONS

**Gaze-to-text Generation: Beyond Categorical Decoding of Human Attention**

**ECCV 2026**

**Sounak Mondal**, D. Samaras, G. Zelinsky, M. Hoai

**Gaze-Language Alignment for Zero-Shot Prediction of Visual Search Targets from Human Gaze Scanpaths**

**ICCV 2025**

**Sounak Mondal**, N. Sendhilnathan, T. Zhang, Y. Liu, M. Proulx, M.L. Iuzzolino, C. Qin, T.R. Jonker

**Look Hear: Gaze Prediction for Speech-directed Human Attention**

**ECCV 2024**

**Sounak Mondal**, S. Ahn, Z. Yang, N. Balasubramanian, D. Samaras, G. Zelinsky, M. Hoai

**Gazeformer: Scalable, Effective and Fast Prediction of Goal-Directed Human Attention**

**CVPR 2023**

**Sounak Mondal**, Z. Yang, S. Ahn, D. Samaras, G. Zelinsky, M. Hoai

|                                                                                                   |                   |
|---------------------------------------------------------------------------------------------------|-------------------|
| <b>Personalized Image Descriptions from Attention Sequences</b>                                   | <b>CVPR 2026</b>  |
| R. Xue, J. Xu, H. Le, <b>Sounak Mondal</b> , A. Leite, G. Zelinsky, M. Hoai, D. Samaras           |                   |
| <b>Few-shot Personalized Scanpath Prediction</b>                                                  | <b>CVPR 2025</b>  |
| R. Xue, J. Xu, <b>Sounak Mondal</b> , H. Le, G. Zelinsky, M. Hoai, D. Samaras                     |                   |
| <b>Unifying Top-down and Bottom-up Scanpath Prediction using Transformers</b>                     | <b>CVPR 2024</b>  |
| Z. Yang, <b>Sounak Mondal</b> , S. Ahn, R. Xue, G. Zelinsky, M. Hoai, D. Samaras                  |                   |
| <b>Target-absent Human Attention</b>                                                              | <b>ECCV 2022</b>  |
| Z. Yang, <b>Sounak Mondal</b> , S. Ahn, G. Zelinsky, M. Hoai, D. Samaras                          |                   |
| <b>Diffusion-Refined VQA Annotations for Semi-Supervised Gaze Following</b>                       | <b>ECCV 2024</b>  |
| Q. Miao, A. Graikos, J. Zhang, <b>Sounak Mondal</b> , M. Hoai, D. Samaras                         |                   |
| <b>Characterizing Target-absent Human Attention</b>                                               | <b>CVPRW 2022</b> |
| Y. Chen, Z. Yang, S. Chakraborty, <b>Sounak Mondal</b> , S. Ahn, D. Samaras, M. Hoai, G. Zelinsky |                   |

## SELECTED AWARDS

---

- ICCV 2025 Doctoral Consortium
- CVPR 2023 Travel Grant Award
- ACM Best B.E. Thesis Award, 2017

## ACADEMIC & TECHNICAL DETAILS

---

**Graduate Courses:** Computer Vision, Natural Language Processing, Robotics, Machine Learning, Database Systems

**Languages & Frameworks:** Python, PyTorch, TensorFlow, C++, C, Java, Hadoop (familiar), Spark (familiar)

**Service:** Reviewer for CVPR, ICCV, ECCV, ICLR, NeurIPS, TPAMI, Nature, BMVC